

CA2321: Do not deserialize with JavaScriptSerializer using a SimpleTypeResolver (code analysis)

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Content

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CA2321: Do not deserialize with JavaScriptSerializer using a SimpleTypeResolver

Summarize this article for me

Property	Value	Rule ID	CA2321	Title	Do not deserialize with JavaScriptSerializer using a SimpleTypeResolver	Category	Security	Fix is breaking or non-breaking	Non-breaking	Enabled by default in .NET 10	No	Applicable languages	C# and Visual Basic
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Cause

A [System.Web.Script.Serialization.JavaScriptSerializer](#) deserialization method was called or referenced after initializing with a [System.Web.Script.Serialization.SimpleTypeResolver](#).

By default, this rule analyzes the entire codebase, but this is [configurable]().

Rule description

Insecure deserializers are vulnerable when deserializing untrusted data. An attacker could modify the serialized data to include unexpected types to inject objects with malicious side effects. An attack against an insecure deserializer could, for example, execute commands on the underlying operating system, communicate over the network, or delete files.

This rule finds [System.Web.Script.Serialization.JavaScriptSerializer](#) deserialization method calls or references, after initializing the [JavaScriptSerializer](#) with a [System.Web.Script.Serialization.SimpleTypeResolver](#).

How to fix violations

- Don't initialize [JavaScriptTypeResolver](#) with a [System.Web.Script.Serialization.SimpleTypeResolver](#).
- If your code needs to read data serialized using a [SimpleTypeResolver](#), restrict deserialized types to an expected list by implementing a custom [JavaScriptTypeResolver](#).
- Make the serialized data tamper-proof. After serialization, cryptographically sign the serialized data. Before deserialization, validate the cryptographic signature. Protect the cryptographic key from being disclosed and design for key rotations.

When to suppress warnings

It's safe to suppress a warning from this rule if:

- You know the input is trusted. Consider that your application's trust boundary and data flows may change over time.
- You've taken one of the precautions in [How to fix violations]().

Suppress a warning

If you just want to suppress a single violation, add preprocessor directives to your source file to disable and then re-enable the rule.

```
#pragma warning disable CA2321
// The code that's violating the rule is on this line.
#pragma warning restore CA2321
```

To disable the rule for a file, folder, or project, set its severity to `none` in the [configuration file](#).

```
[*.{cs,vb}]
dotnet_diagnostic.CA2321.severity = none
```

For more information, see [How to suppress code analysis warnings](#).

Configure code to analyze

Use the following options to configure which parts of your codebase to run this rule on.

- [Exclude specific symbols]()
- [Exclude specific types and their derived types]()

You can configure these options for just this rule, for all rules they apply to, or for all rules in this category ([Security](#)) that they apply to. For more information, see [Code quality rule configuration options](#).

Exclude specific symbols

You can exclude specific symbols, such as types and methods, from analysis by setting the [excluded_symbol_names](#) option. For example, to specify that the rule should not run on any code within types named `MyType`, add the following key-value pair to an `.editorconfig` file in your project:

```
dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType
```

Note

Replace the `XXXX` part of `CAXXXX` with the ID of the applicable rule.

Allowed symbol name formats in the option value (separated by `|`):

- Symbol name only (includes all symbols with the name, regardless of the containing type or namespace).
- Fully qualified names in the symbol's [documentation ID format](#). Each symbol name requires a symbol-kind prefix, such as `M:` for methods, `T:` for types, and `N:` for namespaces.
- `.ctor` for constructors and `.cctor` for static constructors.

Examples:

Option Value	Summary
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType</code>	Matches all symbols named <code>MyType</code> .
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = MyType1 MyType2</code>	Matches all symbols named either <code>MyType1</code> or <code>MyType2</code> .
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = M:NS.MyType.MyMethod(ParamType)</code>	Matches specific method <code>MyMethod</code> with the specified fully qualified signature.
<code>dotnet_code_quality.CAXXXX.excluded_symbol_names = M:NS1.MyType1.MyMethod1(ParamType) M:NS2.MyType2.MyMethod2(ParamType)</code>	Matches specific methods <code>MyMethod1</code> and <code>MyMethod2</code> with the respective fully qualified signatures.

Exclude specific types and their derived types

You can exclude specific types and their derived types from analysis by setting the [exclude_d_type_names_with_derived_types](#) option. For example, to specify that the rule should not

run on any methods within types named `MyType` and their derived types, add the following key-value pair to an `.editorconfig` file in your project:

```
dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType
```

Note

Replace the `XXXX` part of `CAXXXX` with the ID of the applicable rule.

Allowed symbol name formats in the option value (separated by `|`):

- Type name only (includes all types with the name, regardless of the containing type or namespace).
- Fully qualified names in the symbol's [documentation ID format](#), with an optional `T:` prefix.

Examples:

Option value	Summary
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType</code>	Matches all types named <code>MyType</code> and all of their derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = MyType1 MyType2</code>	Matches all types named either <code>MyType1</code> or <code>MyType2</code> and all of their derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = M:NS.MyType</code>	Matches specific type <code>MyType</code> with given fully qualified name and all of its derived types.
<code>dotnet_code_quality.CAXXXX.excluded_type_names_with_derived_types = M:NS1.MyType1 M:NS2.MyType2</code>	Matches specific types <code>MyType1</code> and <code>MyType2</code> with the respective fully qualified names, and all of their derived types.

Pseudo-code examples

Violation 1

```
using System.Web.Script.Serialization;

public class ExampleClass
{
    public T Deserialize<T>(string str)
    {
        JavaScriptSerializer s = new JavaScriptSerializer(new SimpleTypeResolver());
        return s.Deserialize<T>(str);
    }
}
```

```
Imports System.Web.Script.Serialization
```

```
Public Class ExampleClass
```

```
    Public Function Deserialize(Of T)(str As String) As T
```

```
        Dim s As JavaScriptSerializer = New JavaScriptSerializer(New  
SimpleTypeResolver())
```

```
        Return s.Deserialize(Of T)(str)
```

```
    End Function
```

```
End Class
```

Solution 1

```
using System.Web.Script.Serialization;
```

```
public class ExampleClass
```

```
{
```

```
    public T Deserialize<T>(string str)
```

```
    {
```

```
        JavaScriptSerializer s = new JavaScriptSerializer();
```

```
        return s.Deserialize<T>(str);
```

```
    }
```

```
}
```

```
Imports System.Web.Script.Serialization
```

```
Public Class ExampleClass
```

```
    Public Function Deserialize(Of T)(str As String) As T
```

```
        Dim s As JavaScriptSerializer = New JavaScriptSerializer()
```

```
        Return s.Deserialize(Of T)(str)
```

```
    End Function
```

```
End Class
```

Violation 2

```
using System.Web.Script.Serialization;

public class BookRecord
{
    public string Title { get; set; }
    public string Author { get; set; }
    public int PageCount { get; set; }
    public AisleLocation Location { get; set; }
}

public class AisleLocation
{
    public char Aisle { get; set; }
    public byte Shelf { get; set; }
}

public class ExampleClass
{
    public BookRecord DeserializeBookRecord(string s)
    {
        JavaScriptSerializer serializer = new JavaScriptSerializer(new
SimpleTypeResolver());
        return serializer.Deserialize<BookRecord>(s);
    }
}
```

```
Imports System.Web.Script.Serialization
```

```
Public Class BookRecord
```

```
    Public Property Title As String
```

```
    Public Property Author As String
```

```
    Public Property Location As AisleLocation
```

```
End Class
```

```
Public Class AisleLocation
```

```
    Public Property Aisle As Char
```

```
    Public Property Shelf As Byte
```

```
End Class
```

```
Public Class ExampleClass
```

```
    Public Function DeserializeBookRecord(str As String) As BookRecord
```

```
        Dim serializer As JavaScriptSerializer = New JavaScriptSerializer(New  
SimpleTypeResolver())
```

```
        Return serializer.Deserialize(Of BookRecord)(str)
```

```
    End Function
```

```
End Class
```

Solution 2

```

using System;
using System.Web.Script.Serialization;

public class BookRecordTypeResolver : JavaScriptTypeResolver
{
    // For compatibility with data serialized with a JavaScriptSerializer initialized
with SimpleTypeResolver.
    private static readonly SimpleTypeResolver Simple = new SimpleTypeResolver();

    public override Type ResolveType(string id)
    {
        // One way to discover expected types is through testing deserialization
        // of **valid** data and logging the types used.

        ////Console.WriteLine($"ResolveType('{id}')");

        if (id == typeof(BookRecord).AssemblyQualifiedName || id ==
typeof(AisleLocation).AssemblyQualifiedName)
        {
            return Simple.ResolveType(id);
        }
        else
        {
            throw new ArgumentException("Unexpected type ID", nameof(id));
        }
    }

    public override string ResolveTypeId(Type type)
    {
        return Simple.ResolveTypeId(type);
    }
}

public class BookRecord
{
    public string Title { get; set; }
    public string Author { get; set; }
    public int PageCount { get; set; }
    public AisleLocation Location { get; set; }
}

public class AisleLocation
{
    public char Aisle { get; set; }
}

```

```
    public byte Shelf { get; set; }  
}  
  
public class ExampleClass  
{  
    public BookRecord DeserializeBookRecord(string s)  
    {  
        JavaScriptSerializer serializer = new JavaScriptSerializer(new  
BookRecordTypeResolver());  
        return serializer.Deserialize<BookRecord>(s);  
    }  
}
```

```

Imports System
Imports System.Web.Script.Serialization

Public Class BookRecordTypeResolver
    Inherits JavaScriptTypeResolver

    ' For compatibility with data serialized with a JavaScriptSerializer initialized with
    SimpleTypeResolver.
    Private Dim Simple As SimpleTypeResolver = New SimpleTypeResolver()

    Public Overrides Function ResolveType(id As String) As Type
        ' One way to discover expected types is through testing deserialization
        ' of **valid** data and logging the types used.

        ''Console.WriteLine($"ResolveType('{id}')"")

        If id = GetType(BookRecord).AssemblyQualifiedName Or id =
GetType(AisleLocation).AssemblyQualifiedName Then
            Return Simple.ResolveType(id)
        Else
            Throw New ArgumentException("Unexpected type", NameOf(id))
        End If
    End Function

    Public Overrides Function ResolveTypeId(type As Type) As String
        Return Simple.ResolveTypeId(type)
    End Function
End Class

Public Class BookRecord
    Public Property Title As String
    Public Property Author As String
    Public Property Location As AisleLocation
End Class

Public Class AisleLocation
    Public Property Aisle As Char
    Public Property Shelf As Byte
End Class

Public Class ExampleClass
    Public Function DeserializeBookRecord(str As String) As BookRecord
        Dim serializer As JavaScriptSerializer = New JavaScriptSerializer(New
BookRecordTypeResolver())
    
```

```
    Return serializer.Deserialize(Of BookRecord)(str)
End Function
End Class
```

CA2322: Ensure JavaScriptSerializer is not initialized with SimpleTypeResolver before deserializing

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